



Batterjee Medical College
For Science and Technology

Approach to a case of Fever and Palpitation

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Infective endocarditis

Learning Objectives

Knowledge & understanding

Describe the cardiac and extracardiac clinical manifestations of infective endocarditis.

Identify the cardiac conditions and patient risk categories (high, moderate, low) that warrant infective endocarditis prophylaxis.

Skill

Apply the modified Duke criteria to determine the diagnostic likelihood of infective endocarditis.

Formulate a comprehensive management plan for a patient with infective endocarditis who develops a complication

Main Subtopics

- Etiology and Microbiology
- Risk Factors and Risk Classification
- Pathophysiology
- Clinical Features
- Classification
- Diagnostics
- Differential Diagnosis
- Treatment
- Prevention

Case scenario :

A 72-year-old man comes to the physician for medical clearance for a molar extraction. He feels well. He reports he is able to climb 3 flights of stairs without experiencing any shortness of breath. He has hypertension, type 2 diabetes mellitus, and ischemic heart disease. He underwent an aortic valve replacement for severe aortic stenosis last year. Twelve years ago, he underwent a cardiac angioplasty and had 2 stents placed. Current medications include aspirin, warfarin, lisinopril, metformin, sitagliptin, and simvastatin. His temperature is 37.1°C (98.8°F), pulse is 92/min, and blood pressure is 136/82 mm Hg. A grade 6/6 systolic ejection click is heard at the right second intercostal space.

Case problem :

Which of the following is the most appropriate next step in management?

- A. Administer oral amoxicillin 1 hour before the procedure
- B. Obtain echocardiography prior to procedure
- C. Avoid nitrous oxide during the procedure
- D. Discontinue aspirin and warfarin 72 hours prior to procedure
- E. Administer oral azithromycin 1 hour before and 2 hours after the procedure

Infective endocarditis

Infection of the endocardium that typically affects one or more heart valves.

Staphylococcus aureus:

Most common cause of acute IE, including :

- ✓ Persons who inject drugs
- ✓ Patients with prosthetic valves
- ✓ Pacemakers/ICDs

Viridans streptococci

Most common cause of subacute IE, especially following dental procedures

Staphylococcus epidermidis

Bacteremia from infected peripheral venous catheters

Enterococci (especially Enterococcus faecalis)

Common cause of IE following nosocomial UTIs

Streptococcus gallolyticus subsp. gallolyticus (Sgg)

Associated with colorectal cancer

Gram-negative HACEK group

In patients with poor dental hygiene and or periodontal infection

***H**aemophilus*

***A**ggregatibacter*

***C**ardiobacterium*

***E**ikenella*

***K**ingella*

Fungal endocarditis (Candida, Aspergillus fumigatus) At risk groups

- Immunosuppressed patients (e.g., patients with HIV or organ transplantation)
- Persons who inject drugs
- Patients who have received cardiosurgical interventions
- Patients with long-term indwelling IV catheters

Coxiella burnetii Bartonella species

Gram-negative pathogens responsible for culture-negative endocarditis

Etiology - Risk factors for infective endocarditis

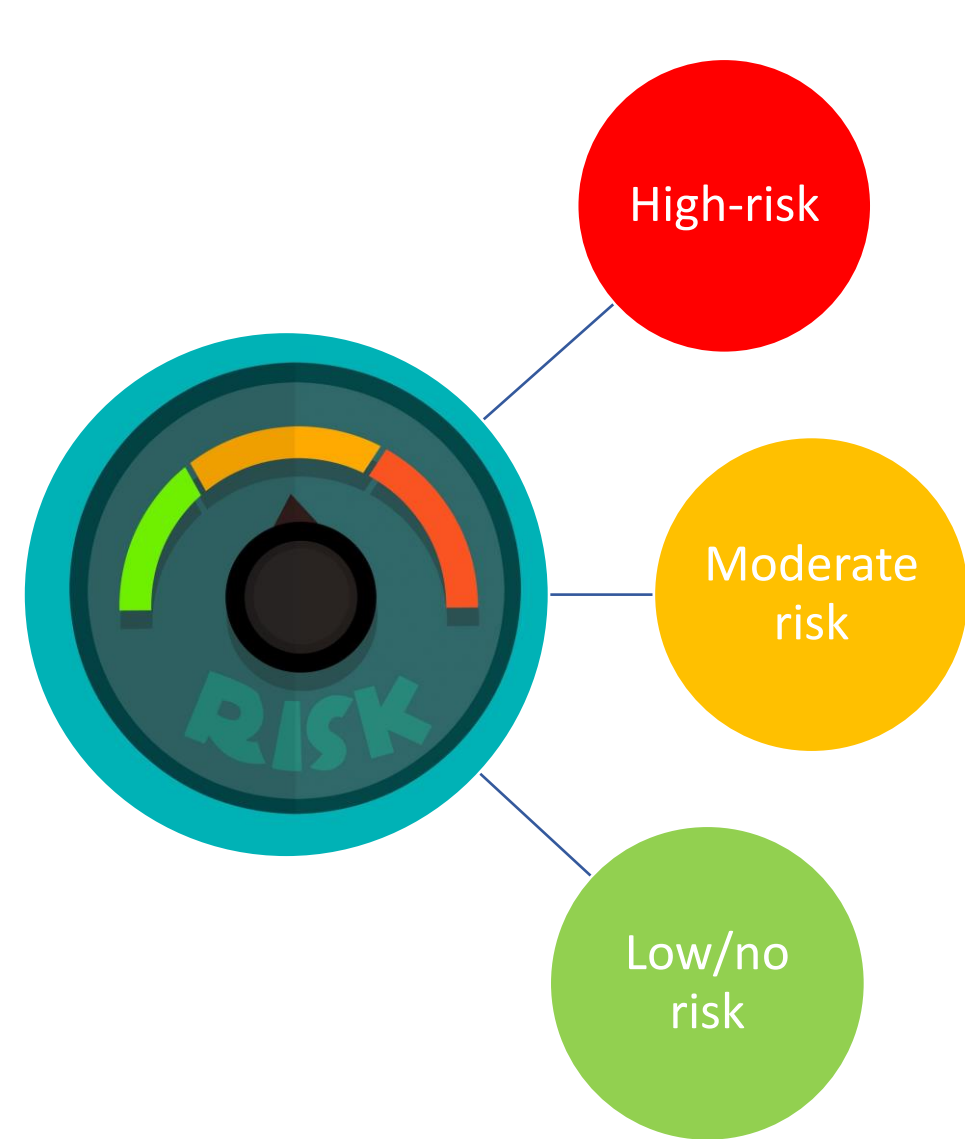
Cardiac conditions

- Acquired valvular disease
- Prosthetic heart valves
- Congenital heart defects

Non-cardiac risk factors

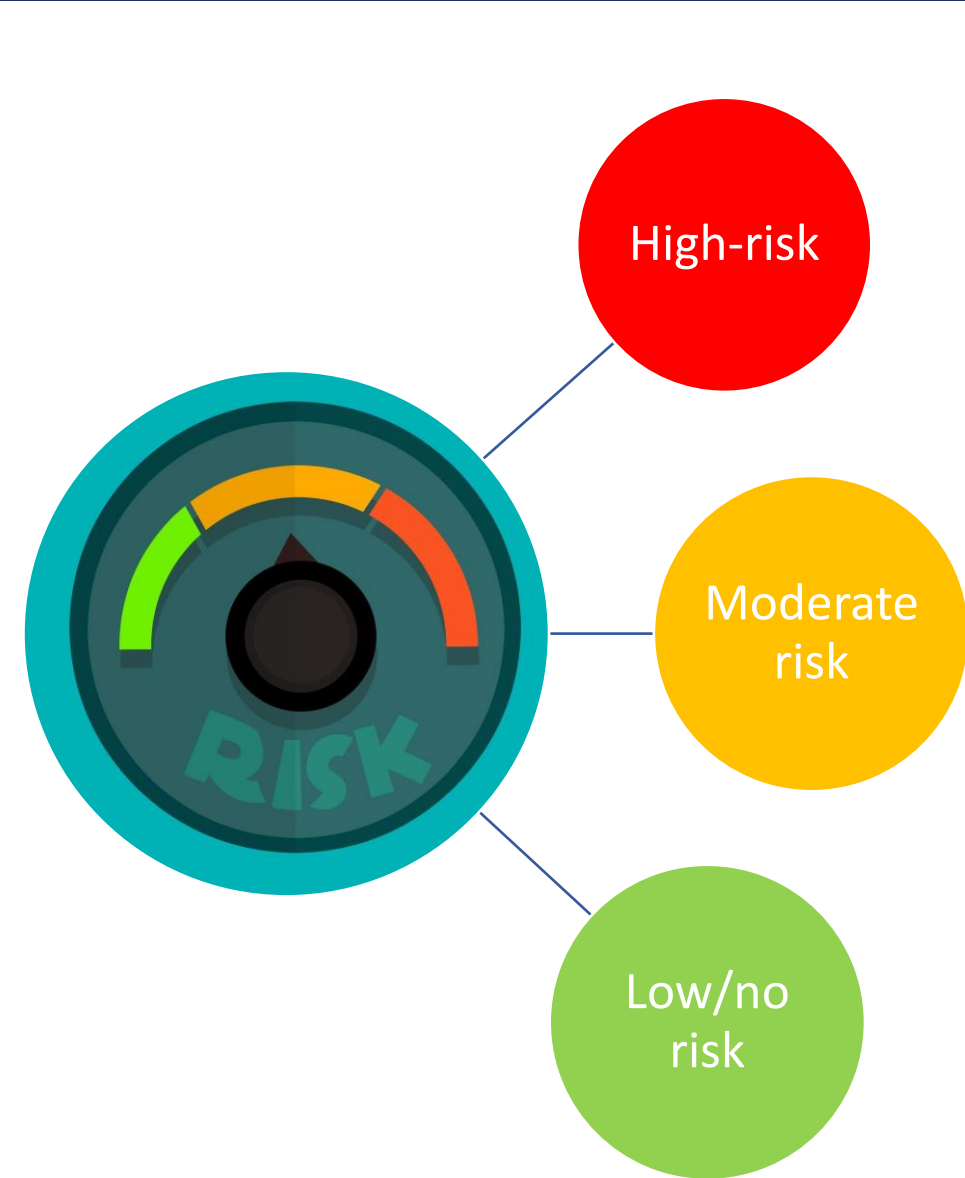
- Poor dental status
- Dental procedures
- Nonsterile venous injections (e.g., Intra Venous Drug Use IVDU)
- Intravascular devices
- Surgery
- HIV
- Hemodialysis

Risk Classification



- Prosthetic cardiac valve
- Previous infective endocarditis (IE)
- Congenital heart disease (unrepaired, repaired within 6 mo, or repaired with defects)
- Cardiac transplant with valve disease (surgically constructed systemic-to-pulmonary shunts or conduits)

Risk Classification



Moderate
risk

- Other congenital cardiac defects,
- Acquired valvular dysfunction,
- Hypertrophic cardiomyopathy

Low/no risk

- Secundum atrial septal defect (ASD)
- Surgically repaired ASD or ventricular septal defect (VSD)
- Patent ductus arteriosus (PDA)
- Mitral valve (MV) prolapse
- Ischemic heart disease
- Previous coronary artery bypass graft (CABG)

Pathophysiology

Damaged valvular endothelium

sterile vegetation

→ bacterial colonization of vegetation

→ valve destruction with loss of function

(**valve regurgitation**)

Valve involvement

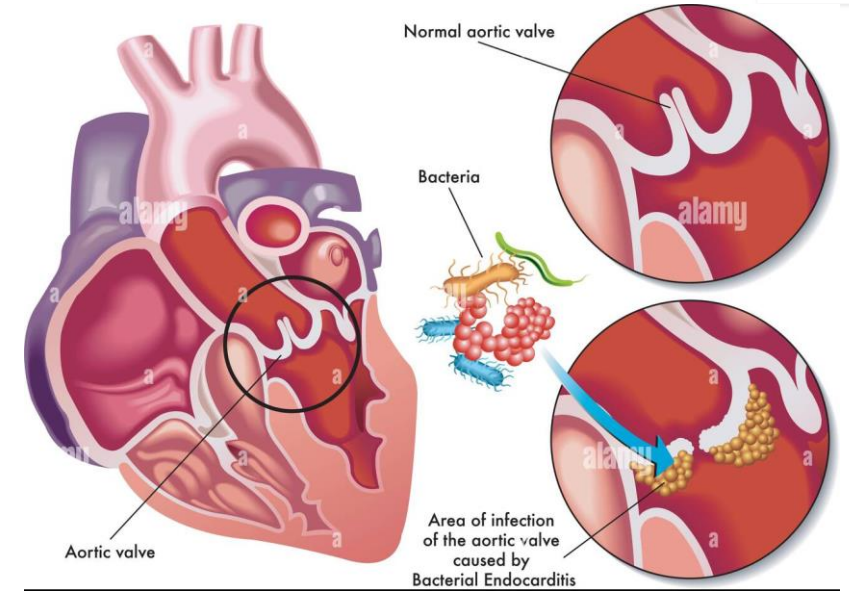
Frequency of valve involvement:

mitral valve > aortic valve > tricuspid valve > pulmonary valve

The tricuspid valve is the most commonly affected valve in persons who inject drugs (associated with *Pseudomonas*, *S. aureus*, and *Candida*).

Clinical consequences

Bacterial vegetation → bacterial thromboemboli → vessel occlusion → infarctions → Emboli can lead to metastatic infections of other organs.



- **Constitutional symptoms**
 - Fever and chills
 - Tachycardia
 - General malaise, weakness, weight loss, night sweats
 - Dyspnea
- **Cardiac manifestations**
- **Extra cardiac manifestations of IE**

Cardiac manifestations

- Development of a new heart murmur **or** change in a preexisting murmur
- Tricuspid valve regurgitation in persons with:
 - ✓ Drugs injection
 - ✓ Immunocompromised individuals
 - ✓ Congenital heart disease, and
 - ✓ Instrumentation in the right heart (e.g., central venous catheters)
- Aortic valve regurgitation
- Mitral valve regurgitation
- Heart failure
- Arrhythmias: Suspect a perivalvular **abscess** in patients with IE who develop a new conduction abnormality (e.g., **heart block**).

Extra cardiac manifestations of IE

Peripheral embolic and immunologic phenomena:

only 5–10% of patients.

Petechiae, especially **splinter hemorrhages**
(underneath fingernails)

Janeway lesions

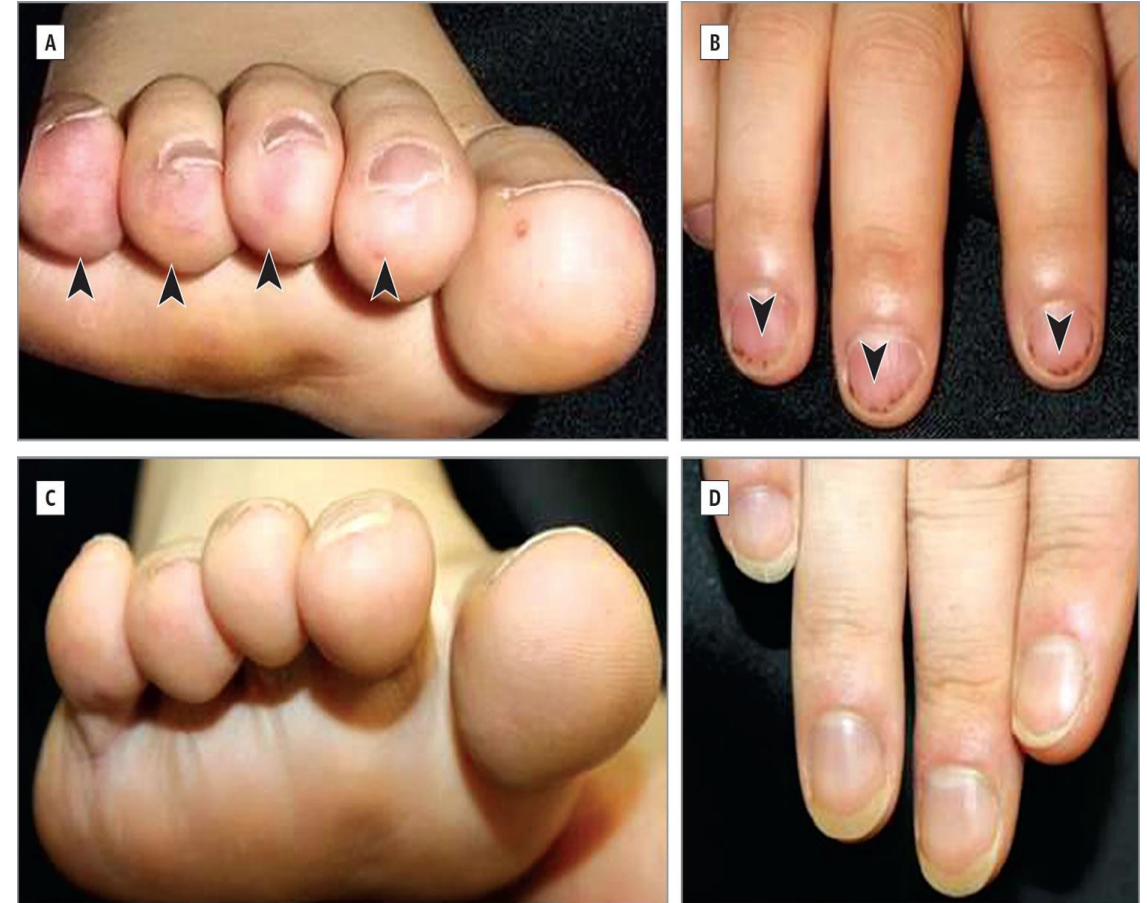
Small, nontender, erythematous macules
(on palms and soles)

Osler nodes:

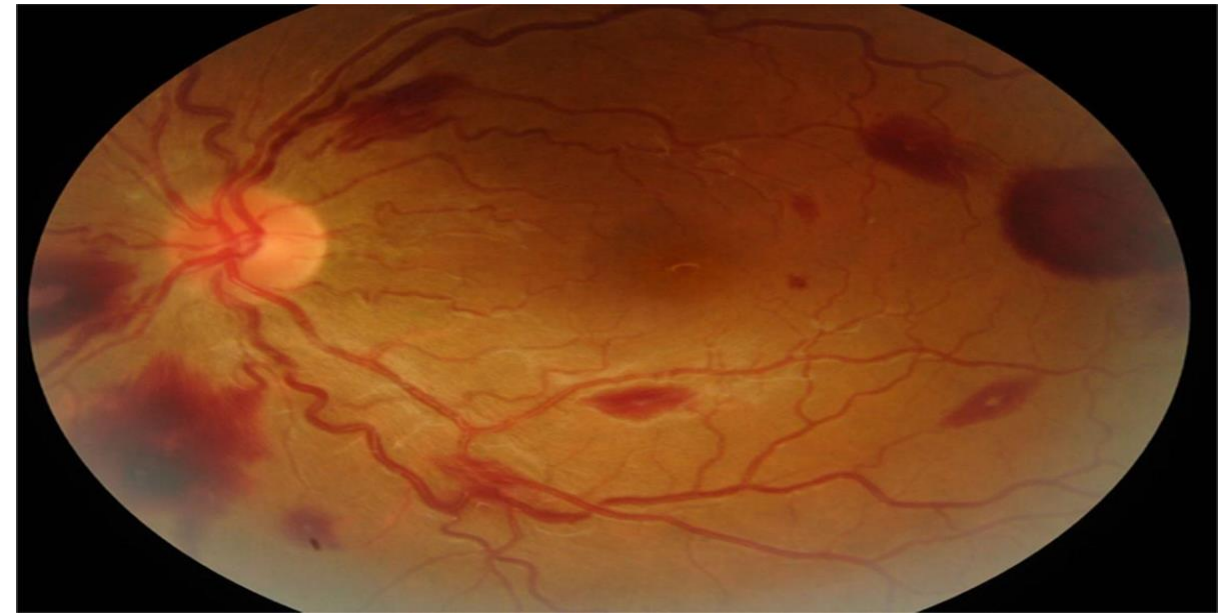
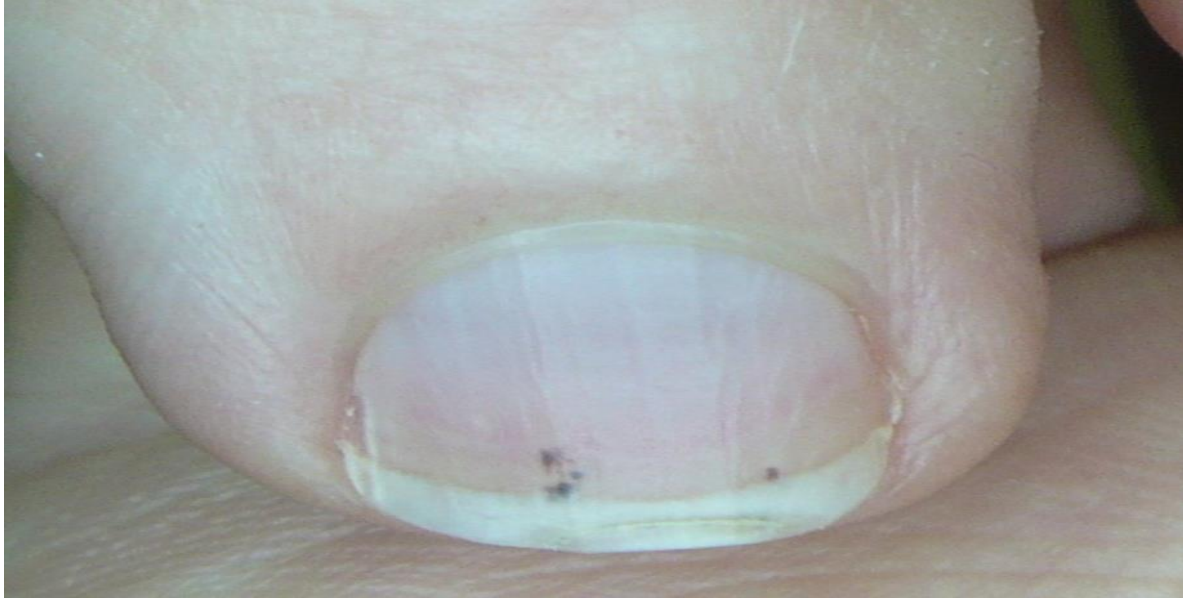
Painful nodules
(on pads of the fingers and toes)

Roth spots:

Round retinal hemorrhages with pale centers



Extra cardiac manifestations of IE



Emboli to intraabdominal organs

- Acute renal injury , including hematuria and anuria,
- Splenomegaly and possible LUQ pain, splenic abscess

Neurological manifestations: septic embolic stroke

Pulmonary manifestations: septic emboli , pulmonary embolism

Others: Arthritis

Summarized clinical picture of Infective Endocarditis

The main clinical picture of infective endocarditis can be summarized using the mnemonic **FROM JANE**, which represents:

Fever

Roth spots

Osler nodes

Murmur

Janeway lesions

Acute glomerulonephritis

Nail-bed hemorrhage

Embolism



Classified by type of valve involved and clinical

	Native valve endocarditis		Prosthetic valve endocarditis
	Acute bacterial endocarditis	Subacute bacterial endocarditis	
Clinical features	• Acute onset • Rapid, fulminant progression (days to weeks) • Severe constitutional symptoms (e.g., high fever)	• Insidious onset • Slow progression (weeks to months) • Less severe constitutional	• Early-onset: ≤ 1 year after surgery • Late-onset: > 1 year after surgery
Main pathogens	• Most common: S. aureus (associated with large vegetations that can destroy the valves) • Others: group A hemolytic streptococci, S.p pneumoniae, N.gonorrhoeae	• Viridans streptococci • S. aureus • Enterococci • HACEK	• Early-onset: Coagulase-negative staphylococci, S. aureus, or gram-negative bacilli (most common) • Late-onset: Coagulase-negative staphylococci, S. aureus, Viridans group streptococci (most common)
Affected valves	• Healthy native valves	• Native valves with prior injury or congenital defects	• Mechanical valves • Bioprosthetic valves

Approach to diagnosis

Suspect IE based on clinical findings (e.g., fever without focus combined with a new murmur) and predisposing conditions.

The modified Duke criteria help categorize the diagnostic likelihood of IE:
definite vs. possible

All patients should receive multiple blood cultures and echocardiography.

MODIFIED DUKE CRITERIA FOR INFECTIVE ENDOCARDITIS

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MNEMONIC : **BE** TIMER

MAJOR CRITERIA

MNEMONIC - **BE**

- **B**lood Culture POSITIVE > 2 times 12 hrs apart.
- **E**chocardiographic evidence of Endocardial Involvement

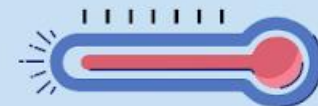


MINOR CRITERIA

MNEMONIC - **TIMER**

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- **T**emperature > 38 ° C (Fever)
- **I**mmunological phenomena (Osler's nodes, Roth spots)
- **M**icrobiological Evidence (Positive blood culture not meeting a major criterion)
- **E**mbolic Phenomenon
(Arterial emboli, septic emboli, conjunctival hemorrhage & painless skin lesions.)
- **R**isk Factors - (Congenital heart condition or IV drug use)



- **DEFINITIVE DIAGNOSIS** : Two major criteria OR one major and three minor criteria OR Five minor criteria.
- **POSSIBLE DIAGNOSIS**: One major and one minor criteria OR Three minor criteria

Laboratory studies

Routine studies

CBC: leukocytosis with left shift

Inflammatory markers: ↑ CRP, ↑ ESR

Echocardiography

Transthoracic echocardiography (TTE) is the initial test of choice for all patients with suspected IE.

- Valvular vegetations
- New valvular regurgitation

Transesophageal echocardiography (TEE) indicated if TTE is non-diagnostic in patients with at least possible endocarditis

Differential diagnoses

- Noninfective endocarditis (nonbacterial thrombotic endocarditis)
- Libman-Sacks endocarditis: verrucous vegetations in SLE or APS
- Prosthetic valve thrombosis

Antibiotics

Common empiric antibiotic regimens

Native valve endocarditis: vancomycin PLUS a beta-lactam

Prosthetic valve endocarditis: Add gentamicin PLUS rifampin

Common targeted antibiotic regimens

Staphylococci

- Methicillin-resistant staphylococci (MRSA): vancomycin
- Prosthetic valve endocarditis; Add gentamicin PLUS rifampin to the regimen described for native valve IE

Viridans group streptococci; penicillin G

HACEK: ceftriaxone (first-line)

Prognosis and mortality

- Adverse prognostic factors:

- CHF
- Prosthetic valve infection
- Valvular/myocardial abscess
- Embolization
- Persistent bacteremia
- Altered mental status

- Mortality:

- Prosthetic valve IE (25-50%),
- Non-IVDU *S. aureus* IE (30-45%),
- IVDU *S. aureus* or streptococcal IE (10-15%)

Endocarditis prophylaxis

Patients with high-risk cardiac features

Procedures requiring IE prophylaxis in patients at risk for IE

- Some dental procedures including tooth extraction
- Placement of a cardiac implantable electronic device (CIED)

Common regimens

(usually administered 30–60 minutes prior to the procedure)

Prior to dental procedures

No penicillin allergy: **amoxicillin**

Penicillin allergy: a **macrolide** (e.g., azithromycin;) OR doxycycline

Recommendations for IE Prophylaxis

1. Prosthetic cardiac valves
2. Prosthetic material used for cardiac valve repair
3. Previous IE.
4. Unrepaired cyanotic congenital heart disease or repaired congenital heart disease, with residual shunts or valvular regurgitation at the site of or adjacent to the site of a prosthetic patch or prosthetic device.
5. Cardiac transplant with valve regurgitation attributable to a structurally abnormal valve



References & further reading

- Otto CM, Nishimura RA, Bonow RO, et al. 2020 ACC/AHA Guideline for the Management of Patients With Valvular Heart Disease: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. Circulation. 2020; 143(5). doi: 10.1161/cir.0000000000000923
- AMBOSS
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Thank You

Questions

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